

Real-time American Sign Language Alphabet (ASL) recognition using Convolutional Neural Networks (CNNs)

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Abstract

American Sign Language (ASL) Alphabet recognition is important to bridge the communication gap with deaf people. This project aims to build real-time American Sign Language (ASL) alphabet recognition using two convolutional neural networks (CNNs) trained on publicly available ASL alphabet datasets on Kaggle. The first model is a simple custom CNN named AslNet, and the second model is Resnet18 fine-tuned on ASL Alphabet. Both models are trained on static hand ASL alphabet images and use data augmentation techniques to improve their generalization. The trained models are integrated with OpenCV to do real-time classification on the webcam feed. AslNet achieved a training accuracy of 89.54%, and Resnet18 achieved 99.81%. AslNet achieved an accuracy of 16.78%, while Resnet18 achieved an accuracy of 54.25% on the validation dataset. These results suggest that although the model is learning from the training dataset, It did not generalize well to real-world ASL alphabet images.

1. Introduction

The American Sign Language alphabet is the foundation of the language. It is used by deaf people to finger spell and express their thoughts. However, there is still a communication barrier with deaf people. The implementation of real-time alphabet recognition could significantly reduce this communication barrier. That is why there is a need for an ASL alphabet recognition system that is accurate and capable of doing real-time recognition.

Convolutional Neural Networks have shown amazing results in image recognition tasks, as shown by models like Resnet and Alexnet. CNNs have also been used to recognize the ASL alphabet, achieving accuracy results of over 90% (Triwijoyo et al. 2023). However, these models are usually validated using a controlled dataset similar to the training dataset. This presents a research gap in how well the CNN model performs on real-world images of the ASL alphabet.

This paper presents a comparative study between a simple custom CNN (AslNet) and fine-tuned Resnet18, both validated on real-world test images. These models are also integrated with OpenCV for real-time ASL alphabet recognition.

2. Methodology

This section outlines my approach to training two CNN models for real-time ASL alphabet recognition. This study uses a quantitative experimental approach to compare

the results of the two models. The data gathering, preprocessing, model architecture, training, and real-time implementation process are discussed in detail below.

2.1. Data

The training dataset is a kaggle dataset consisting of 87,000 images of static hand ASL alphabet with an image size of 200x200 pixels (Nagaraj 2018). It has 29 classes consisting of the 26 letters of the alphabet with three additional classes: SPACE, NOTHING, and DELETE.

The validation dataset is another Kaggle dataset consisting of 870 images (Rasband 2018). Its size and classes are similar to the training dataset. This validation dataset is intended to represent images of static hand ASL alphabet in a real-world setting with different backgrounds and lighting. I decided to use this as the validation dataset to see if the model is able to generalize to real-world images.

2.2. Data Preprocessing



Figure 1. Data Augmentation on Training Dataset

The images used retained their three-channel RGB format. The dataset used for AslNet is resized to 64x64 pixels, while the dataset used for Resnet18 is resized to 224x224 pixels, which are the sizes that each model expects.

Data augmentation is a common practice to improve a model's ability to generalize and prevent overfitting. Some studies use data augmentation techniques such as salt and pepper noise, random rotations, and random shifting to improve their model's robustness (Rangal et al. 2024). I have also decided to use a combination of these techniques: random rotation, random horizontal flipping, random zooming and shifting, lighting and color adjustment, and blurring.

The dataset is also converted to tensors and normalized using the mean of [0.485, 0.456, 0.406] and standard deviation of [0.229, 0.224, 0.225], which are the standard values from Imagenet.

The data augmentation is only applied to the training dataset, while the resizing, converting to tensors, and normalization are applied to both the testing and validation datasets.

2.3. Model Architecture

Two models are trained for this project. One is AslNet, which is a simple custom CNN, and the other is Resnet18, which is a pre-trained model and is fine-tuned on the ASL alphabet.



Figure 2. AslNet Model Architecture

The AslNet Model architecture expects an input image of size 64x64 with three input channels (RGB). For the feature extraction stage, the model consists of three convolutional blocks, each consisting of a 3x3 convolutional layer with an increasing number of filters (32, 64, 128). Each convolutional block has a ReLU activation function to introduce nonlinearity, along with 2x2 max pooling. The output of the final pooling is flattened into a one-dimensional vector. It is passed into a dense layer (fully connected) with 512 neurons and ReLU activation, followed by a dropout layer with $p=0.5$ to prevent overfitting. The final output layer is a dense layer of 29 neurons representing the 29 classes from the dataset. This model architecture is inspired by VGG16 architecture featuring multiple convolutional blocks, each with Relu activation and followed by max pooling, and the final layer consisting of dense layers (Varshney 2020).

For the pre-trained model, Resnet18 was selected, as it achieved the highest accuracy from a comparative analysis of CNN architectures on ASL alphabet classification (Fang 2024). It is loaded with the pre-trained weights, and only the final layer was modified to have 29 output neurons to match the number of classes from the dataset. Resnet18 is a modern CNN architecture that utilizes residual connections to train deeper networks without a decrease in performance (Modi 2021).

2.4. Training Procedure

Both models are trained using categorical cross-entropy loss and the Adam optimizer with a learning rate of 0.001. These configurations are chosen as it is the standard used for image classification. AslNet was trained for 20 epochs, while Resnet18 was trained for five epochs. Resnet18 was trained for fewer epochs as it

reached high accuracy on the training dataset and plateaued a lot sooner. The loss and accuracy for both the training and validation datasets were tracked during each epoch to monitor overfitting and were later used as a metric to compare the two models. Both models are trained in Google Colab utilizing the A100 GPU and using a batch size of 1024 to maximize the GPU's RAM. During the training, the model with the best accuracy on the validation dataset was saved and was later used for real-time inference.

2.5. Real-Time Recognition Implementation

A real-time ASL alphabet recognition system was developed to test how the models perform with real-world images via the computer webcam. The trained models are loaded onto the program for inferencing. OpenCV was used to access the computer's webcam feed. A region of interest (ROI) of 500x500 pixels is indicated by a green square. The ROI was extracted, converted to RGB format, and preprocessed identically to the validation dataset to ensure consistency in the input images. It is then passed onto the model for classification, and the ASL letter is displayed on top of the ROI in real time.

3. Results and Discussion

The aim of the paper is to develop a real-time ASL alphabet recognition system and compare the accuracy of the two models, AslNet and Resnet18.

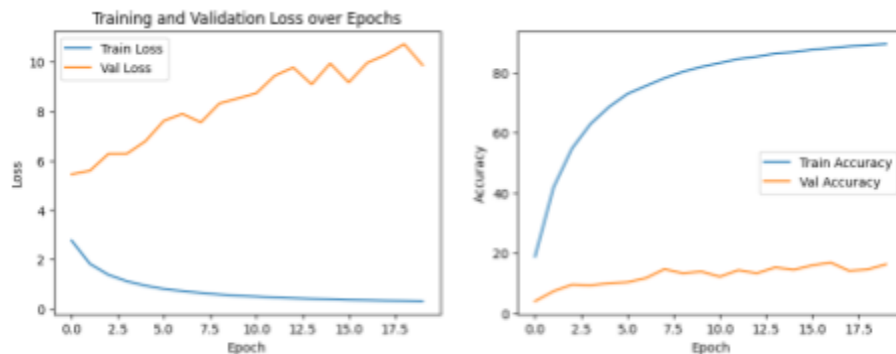


Figure 3. AslNet Loss and Accuracy over Epochs

The AslNet was trained on 20 epochs. The training loss was consistently decreasing and achieving the lowest training loss of 0.2929, while the validation loss was significantly higher and increasing over epochs. The validation loss of the saved model reached 9.9651. The training accuracy increased from 18.78% in the first epoch and reached an accuracy of 89.94 in the 20th epoch. The validation accuracy remained relatively low, with the best accuracy of 16.78%. These results could be attributed to overfitting and the inability of the model to generalize to real-world ASL Alphabet

images. However, these results are expected, as with other custom CNN models achieving a validation accuracy of 48.28% on the same validation set (Little et al. 2020).

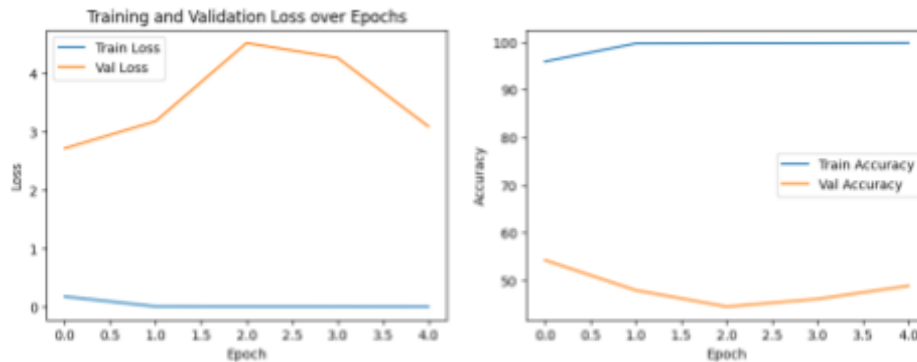


Figure 4. Resnet18 Loss and Accuracy over Epochs

Resnet18 was trained on 5 epochs. It achieved a training loss of 0.0146 during the first epoch and 0.0045 in the last epoch. The validation loss is higher, starting at 2.719 and ending at 3.0940. The training accuracy is 95.92% on the first epoch and hovered around 99% on the following epochs. The validation accuracy was highest during the first epoch at 54.25%, and it hovered around 48% for the following epochs. The high accuracy of Resnet18 during the first couple of epochs is consistent with the result of other analyses of using Resnet for ASL alphabet classification (Agangiba 2023).

Model	Training Loss	Validation Loss	Training Accuracy	Validation Accuracy
AslNet	0.2929	9.9651	88.18%	16.78%
Resnet18	0.0146	2.7195	95.92%	54.25%

Figure 5. Loss and Accuracy comparison between AslNet and Resnet18

During the training process, only the model with the highest validation accuracy was saved for each model. In Figure 5, the metrics for these models are shown. Renest18 showed better accuracy in both the training and validation. It also shows a lower loss on both training and validation. Resnet18 has a better performance compared to AslNet based on these metrics. However, both models suffer from higher validation loss and lower validation accuracy. This poor performance on the validation dataset stems from the model's inability to generalize well on real-world images of the ASL alphabet. It can also be attributed to a data shift problem where there is a mismatch between the training dataset and the validation dataset resulting in lower validation accuracy (Palvel 2023). In this case the training dataset was taken in a

controlled environment, while the validation dataset was taken to mimic the ASL alphabet in real-world settings.

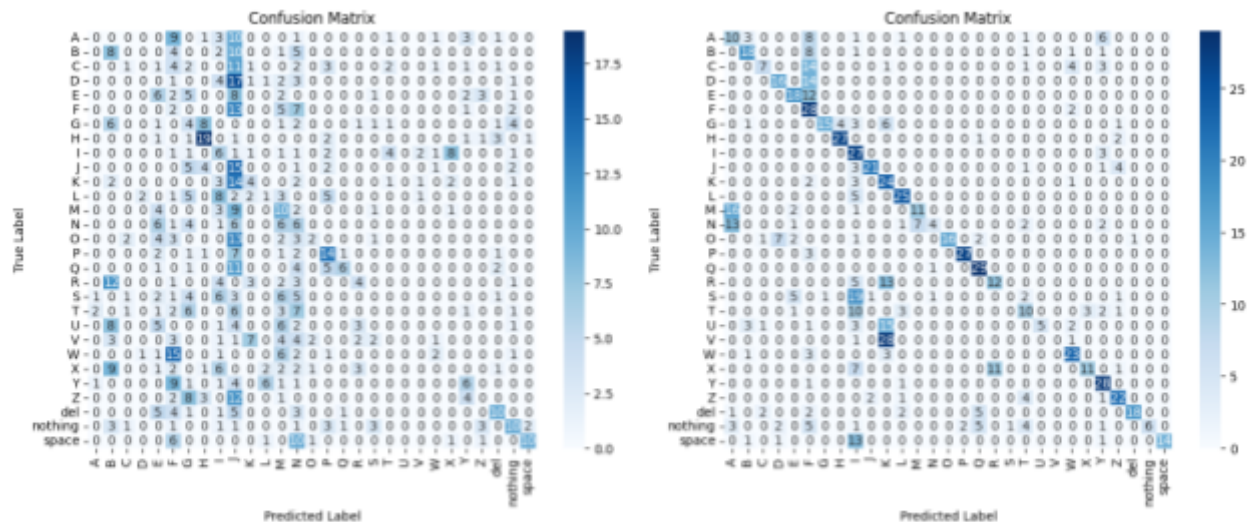


Figure 6. Confusion Matrix of AslNet (left) and Resnet18 (right)

AslNet, with an accuracy of 16.71%, has its confusion matrix on the left side of Figure 6. It shows that AslNet is inaccurate on most of the letters, and it classifies most of the letters as the letter J. AslNet is most accurate in classifying the letter H based on the confusion matrix.

Resnet18, with an accuracy of 54.25%, has its confusion matrix on the right side of Figure 6. It shows that Resnet18 is more accurate in classifying the ASL alphabet. It is most accurate with the letters Q, F, and Y, but has a strong bias with the letter F misclassifying A to E as F. It is inaccurate with classifying the letter V as it classifies it as K 93% of the time.



Figure 7. AslNet real-time recognition

AslNet showed inaccuracies during the real-time recognition. It incorrectly classified A as N, B as E, and V as B. It also showed the same inaccurate classification for the other letters. However, it can correctly classify the 'nothing' class. These results are as expected due to its low validation accuracy.

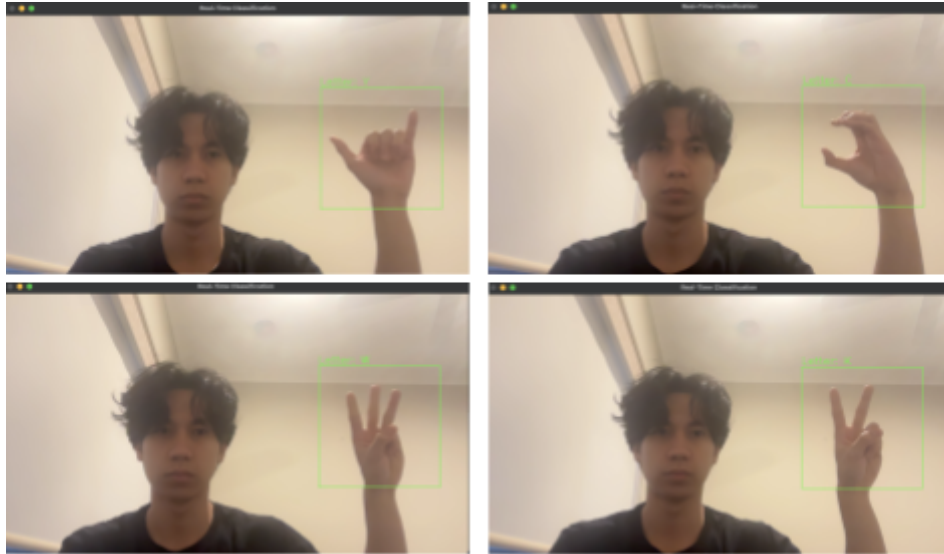


Figure 8. Resnet18 real-time recognition

Resnet18 showed better accuracy during the real-time recognition. It correctly classified the letters Y, C, and W. It is also consistent in classifying the other letters. Similar to the confusion matrix, it also incorrectly classifies V as K. This shows that the resnet18 could be used for real-time ASL alphabet recognition. However, it could be improved to achieve a higher accuracy score.

4. Conclusion

This project aimed to develop a real-time ASL alphabet recognition system and compare two CNN models on classifying real-world ASL alphabet images. Both models achieved good results on the training dataset. However, their performance on the validation dataset is relatively low. The Resnet18 model is the more feasible model for real-time recognition as it can correctly classify the letters, while AslNet showed major inaccuracies. This low validation accuracy highlights a common challenge in the field which is a data shift from custom datasets to real-world images. To further improve the accuracy of both models, improving the diversity of the training dataset, using more advanced data augmentation, and using techniques like hand segmentation could improve the model's performance.

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